

Task 64: 01 Matrix

Problem Description

Given an $m \times n$ binary matrix `mat`, return the distance of the nearest 0 for each cell. The distance between two adjacent cells is 1.

Java Solution

```
class Solution {
    public int[][] updateMatrix(int[][] mat) {
        int m = mat.length, n = mat[0].length;
        Queue<int[]> q = new LinkedList<>();

        for (int i = 0; i < m; i++) {
            for (int j = 0; j < n; j++) {
                if (mat[i][j] == 0) {
                    q.offer(new int[]{i, j});
                } else {
                    mat[i][j] = -1; // unvisited
                }
            }
        }

        int[][] dirs = {{-1, 0}, {1, 0}, {0, -1}, {0, 1}};
        while (!q.isEmpty()) {
            int[] curr = q.poll();
            for (int[] d : dirs) {
                int r = curr[0] + d[0];
                int c = curr[1] + d[1];
                if (r >= 0 && r < m && c >= 0 && c < n && mat[r][c] == -1)
                {
                    mat[r][c] = mat[curr[0]][curr[1]] + 1;
                    q.offer(new int[]{r, c});
                }
            }
        }
        return mat;
    }
}
```